

COMP575

Computational Intelligence

Tutorial 8

MLP Recap

- Input layer
 - Input nodes (not neurons!)
- Hidden layer(s)
 - Neurons (individual units consisting of an activation function which received a weighted sum and a bias term)
 - The number of neurons depends on the size of our dataset
 - No proper scientific way of setting the number, we set them holistically
- Output layers
 - Neurons
 - The number of neurons depends on the number of classes we can predict (classification) OR a single output neuron (regression)
- Each node/neuron is connected to ALL other nodes in the next layer
 - Connections are weighted
 - Weights (and biases) are updated during training via error backpropagation

MLP Recap

- We feed the input into the network in the input layer
- Then we calculate the weighted sum of each neuron
- Repeat until we reach the output layer, where we make our prediction
 - If it's classification, the output neuron with the highest value fires, making the model predict the corresponding label
- We calculate the error and feed it back to the network so the weights (and biases) can be updated so that the network can make a more accurate prediction
 - This is called error backpropagation
- Repeat for a specified number of epochs (one epoch = the model sees the entire training data, NOT one iteration!) or until some kind of convergence criterion is met

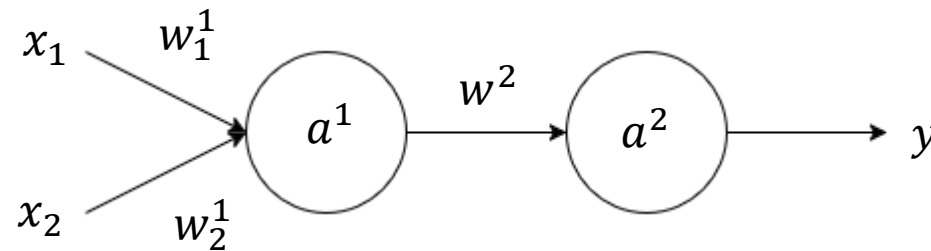
Question 1

- Let's consider a 2-layer neural network
- 2 inputs: x_1 and x_2 with one output y .
 - $x_1 = 1$
 - $x_2 = 2$
- 2 neurons: a^1 and a^2 each with a different activation function.
 - $a^1 = \phi_1(x) = \alpha x$ (linear activation)
 - $a^2 = \phi_2(x) = \frac{1}{1+e^{-x}}$ (sigmoid activation)
- Some weight values...
- The parameter α in the activation function ϕ_1 is $\alpha = 0.5$
- The prediction y is $y = a^2$
- Desired output $d = 1$
- Learning rate $\eta = 0.1$
- Useful tool: https://www.tinkershop.net/ml/sigmoid_calculator.html

Question 1

- $a^1 = \phi_1(\mathbf{x}) = \alpha \mathbf{x}$ (linear activation)
 - $\alpha = 0.5$
- $a^2 = \phi_2(\mathbf{x}) = \frac{1}{1+e^{-x}}$ (sigmoid activation)
- $y = a^2$

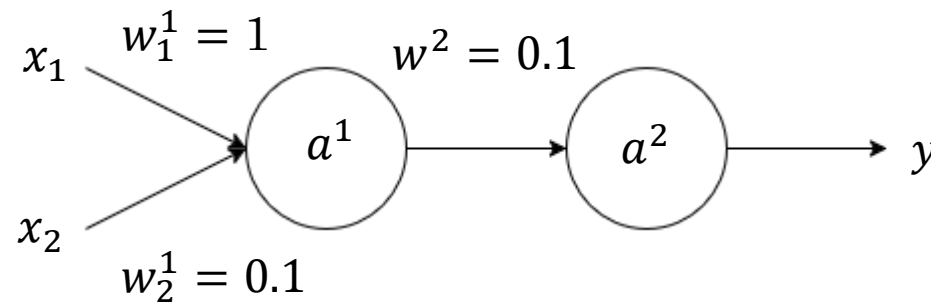
Weight	Value
w_1^1	1
w_2^1	0.1
w^2	0.1



Question 1

- $a^1 = \phi_1(\mathbf{x}) = \alpha \mathbf{x}$ (linear activation)
 - $\alpha = 0.5$
- $a^2 = \phi_2(\mathbf{x}) = \frac{1}{1+e^{-x}}$ (sigmoid activation)
- $y = a^2$

Weight	Value
w_1^1	1
w_2^1	0.1
w^2	0.1



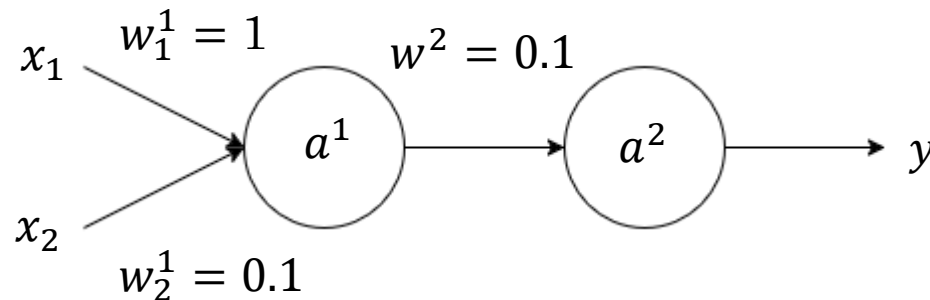
Question 1

- a) Calculate the induced local field and output for each neuron
- b) Compute the gradient of the error with respect to the weights using backprop
- c) Update the weights (use the learning rate (LR))
- d) Calculate the output to the input values x_1 and x_2

Param	Value
x_1	1
x_2	2
d	1
η (LR)	0.1

Weight	Value
w_1^1	1
w_2^1	0.1
w^2	0.1

$$a^1 = \phi_1(x) = 0.5x$$
$$a^2 = \phi_2(x) = \frac{1}{1+e^{-x}}$$
$$y = a^2$$



Question 1

- a) Calculate the induced local field and output for each neuron

Steps:

1. Get the weighted sum

$$x = \sum x_n w_n$$

2. Feed the weighted sum into an activation function

$$f(x) = \phi_n(x)$$

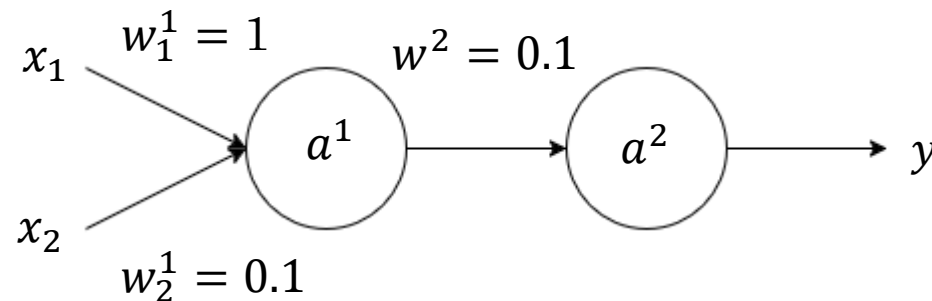
Param	Value
x_1	1
x_2	2
d	1
η (LR)	0.1

Weight	Value
w_1^1	1
w_2^1	0.1
w^2	0.1

$$a^1 = \phi_1(x) = 0.5x$$

$$a^2 = \phi_2(x) = \frac{1}{1+e^{-x}}$$

$$y = a^2$$



Question 1

- a) Calculate the induced local field and output for each neuron

Steps:

1. Get the weighted sum

$$x = \sum x_n w_n$$

$$a^1 = x_1 w_1^1 + x_2 w_2^1$$
$$a^1 = 1 \times 1 + 2 \times 0.1 = 1.2$$

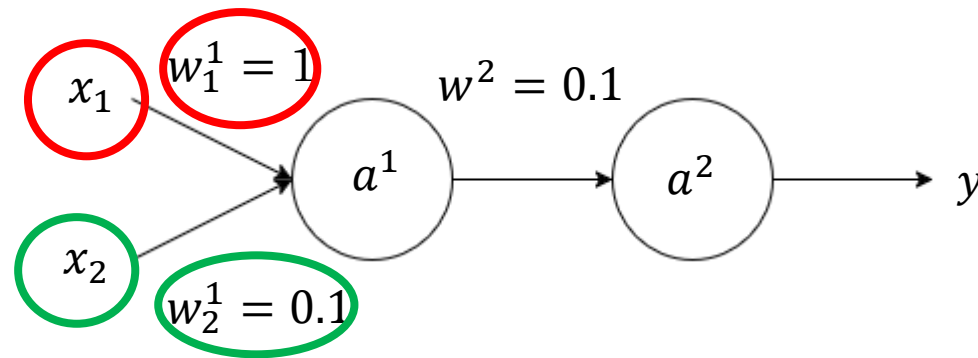
Param	Value
x_1	1
x_2	2
d	1
η (LR)	0.1

Weight	Value
w_1^1	1
w_2^1	0.1
w^2	0.1

$$a^1 = \phi_1(x) = 0.5x$$

$$a^2 = \phi_2(x) = \frac{1}{1+e^{-x}}$$

$$y = a^2$$



Question 1

- a) Calculate the induced local field and output for each neuron

Steps:

1. Get the weighted sum

$$x = \sum x_n w_n$$

$$a^1 = x_1 w_1^1 + x_2 w_2^1$$
$$a^1 = 1 \times 1 + 2 \times 0.1 = 1.2$$

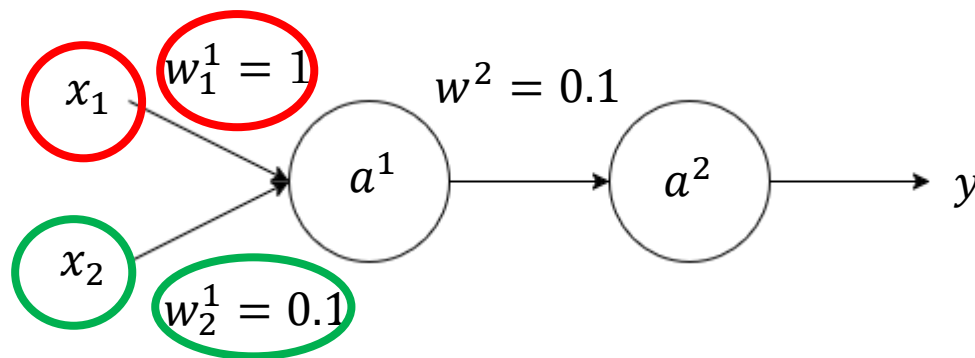
2. Feed the weighted sum into an activation function

$$f(x) = \phi_n(x)$$
$$f(a^1) = 0.5 \times 1.2 = 0.6$$

Param	Value
x_1	1
x_2	2
d	1
η (LR)	0.1
a^1	0.6

Weight	Value
w_1^1	1
w_2^1	0.1
w^2	0.1

$$a^1 = \phi_1(x) = 0.5x$$
$$a^2 = \phi_2(x) = \frac{1}{1+e^{-x}}$$
$$y = a^2$$



Question 1

- a) Calculate the induced local field and output for each neuron

Repeat for neuron

$$a^2 = \sum x_n w_n$$

$$a^2 = 0.6 \times 0.1 = 0.06$$

$$f(x) = \phi_n(x)$$
$$f(a^2) = \frac{1}{1 + e^{-0.06}} = 0.5150$$

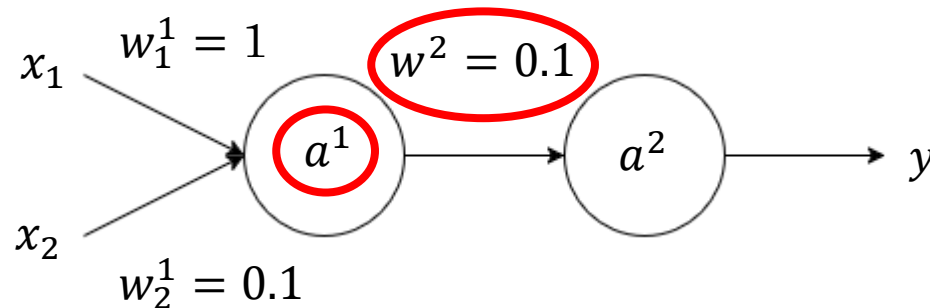
Param	Value
x_1	1
x_2	2
d	1
η (LR)	0.1
a^1	0.6
a^2	0.5150

Weight	Value
w_1^1	1
w_2^1	0.1
w^2	0.1

$$a^1 = \phi_1(x) = 0.5x$$

$$a^2 = \phi_2(x) = \frac{1}{1 + e^{-x}}$$

$$y = a^2$$



Question 1

b) Compute the gradient of the error with respect to the weights using backprop

1. Calculate the derivative (δ_2) of neuron a^2

1. Find the error

$$e_2 = d - y = 1 - 0.5150 = 0.4850$$

2. Find the derivative of the activation function

$$\phi'_2(v_2) = \phi_2(v_2)(1 - \phi_2(v_2))$$

$$\phi'_2(v_2) = 0.5150(1 - 0.5150) = 0.2498$$

3. Put everything together

$$\delta_2 = e_2 \phi'_2(v_2)$$

$$\delta_2 = 0.4850 \times 0.2498 = 0.1212$$

Param	Value
x_1	1
x_2	2
d	1
η (LR)	0.1
a^1	0.6
a^2	0.5150

Weight	Value
w_1^1	1
w_2^1	0.1
w^2	0.1

$$a^1 = \phi_1(x) = 0.5x$$

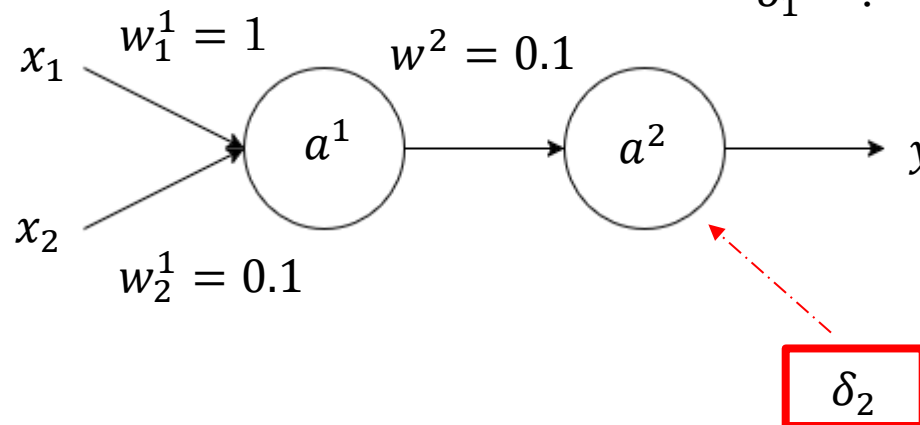
$$a^2 = \phi_2(x) = \frac{1}{1+e^{-x}}$$

$$y = a^2$$

$$d = 1$$

$$\delta_2 = ?$$

$$\delta_1 = ?$$



Question 1

- b) Compute the gradient of the error with respect to the weights using backprop
2. Calculate the derivative (δ_1) of neuron a^1 . Traverse backwards in the network:

$$\delta_1 = \phi'_1(v_2)\delta_2 w^2$$
$$\delta_1 = 0.5 \times 0.1212 \times 0.1 = 0.0061$$

Param	Value
x_1	1
x_2	2
d	1
η (LR)	0.1
a^1	0.6
a^2	0.5150

Weight	Value
w_1^1	1
w_2^1	0.1
w^2	0.1

$$a^1 = \phi_1(x) = 0.5x$$

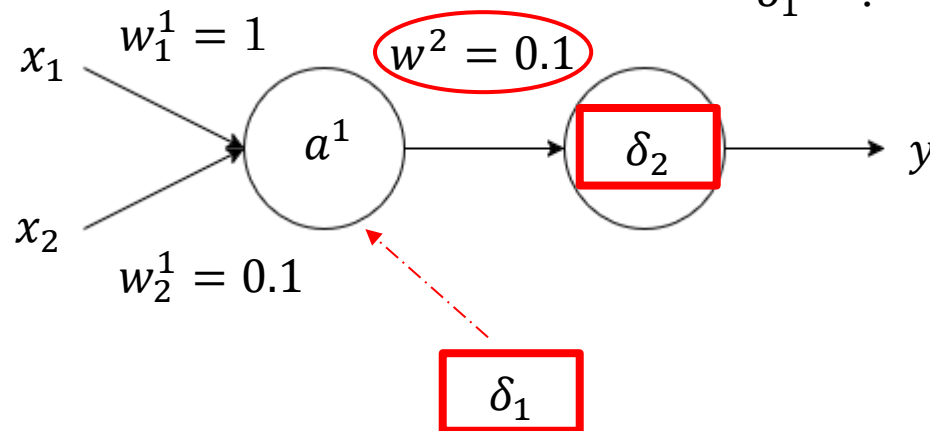
$$a^2 = \phi_2(x) = \frac{1}{1+e^{-x}}$$

$$y = a^2$$

$$d = 1$$

$$\delta_2 = 0.1212$$

$$\delta_1 = ?$$



Question 1

- b) Compute the gradient of the error with respect to the weights using backprop.

Calculate the derivative (δ_1) of neuron a^1 . Traverse backwards in the network:

$$\delta_1 = \phi'_1(v_2)\delta_2w^2$$
$$\delta_1 = 0.5 \times 0.1212 \times 0.1 = 0.0061$$

Param	Value
x_1	1
x_2	2
d	1
η (LR)	0.1
a^1	0.6
a^2	0.5150

Weight	Value
w_1^1	1
w_2^1	0.1
w^2	0.1

$$a^1 = \phi_1(x) = 0.5x$$

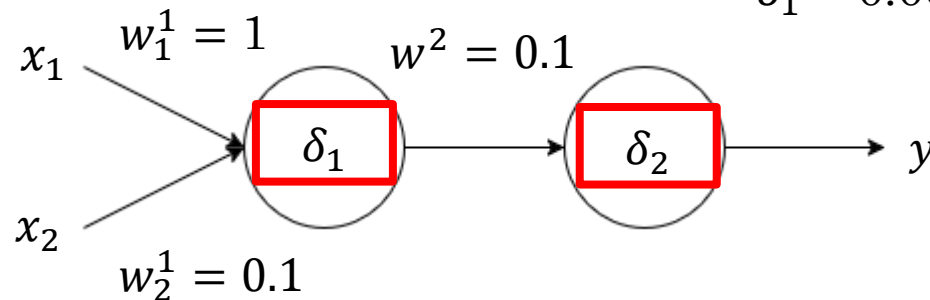
$$a^2 = \phi_2(x) = \frac{1}{1+e^{-x}}$$

$$y = a^2$$

$$d = 1$$

$$\delta_2 = 0.1212$$

$$\delta_1 = 0.0061$$



Question 1

- b) Compute the gradient of the error with respect to the weights using backprop.

Calculate the derivative of the error with respect to each weight.

$$\frac{\partial E}{\partial w^2} = -\delta_2 a^1 = -0.1212 \times 0.6 = -0.0727$$

$$\frac{\partial E}{\partial w_1^1} = -\delta_1 x_1 = -0.0061$$

$$\frac{\partial E}{\partial w_2^1} = -\delta_1 x_2 = -0.0122$$

Param	Value
x_1	1
x_2	2
d	1
η (LR)	0.1
a^1	0.6
a^2	0.5150

Weight	Value
w_1^1	1
w_2^1	0.1
w^2	0.1

$$a^1 = \phi_1(x) = 0.5x$$

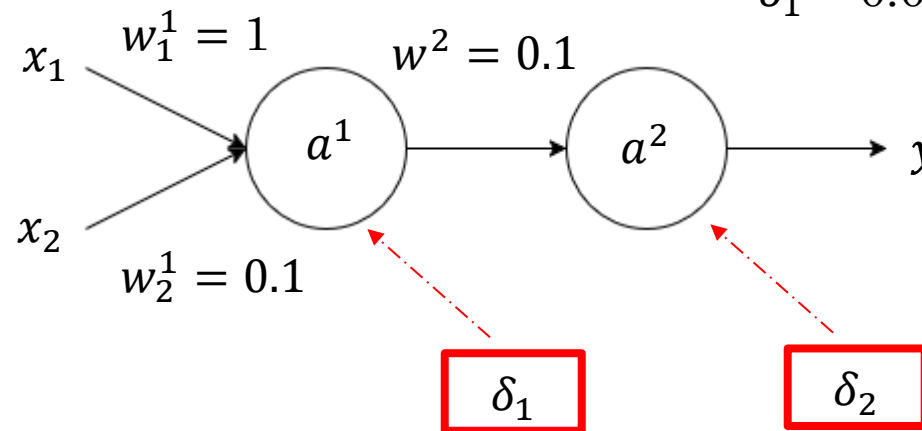
$$a^2 = \phi_2(x) = \frac{1}{1+e^{-x}}$$

$$y = a^2$$

$$d = 1$$

$$\delta_2 = 0.1212$$

$$\delta_1 = 0.0061$$



Question 1

- c) Update the weights (use the learning rate (LR))

$$w_1^{1,+} = w_1^1 - \eta \frac{\partial E}{\partial w_1^1}$$

$$= 1 - 0.1 \times (-0.0061) = 1.0006$$

$$w_2^{1,+} = w_2^1 - \eta \frac{\partial E}{\partial w_2^1} = 0.1012$$

$$w^{2,+} = w^2 - \eta \frac{\partial E}{\partial w^2} = 0.1073$$

Param	Value
x_1	1
x_2	2
d	1
η (LR)	0.1
a^1	0.6
a^2	0.5150

Weight	Value
w_1^1	1
w_2^1	0.1
w^2	0.1

$$a^1 = \phi_1(x) = 0.5x$$

$$a^2 = \phi_2(x) = \frac{1}{1+e^{-x}}$$

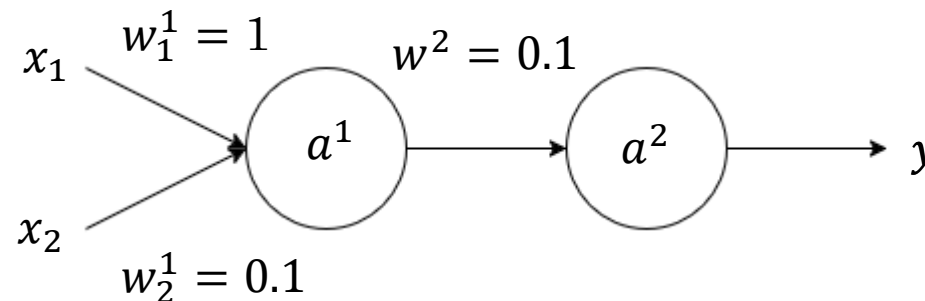
$$y = a^2$$

$$d = 1$$

$$\frac{\partial E}{\partial w^2} = -0.0727$$

$$\frac{\partial E}{\partial w_1^1} = -0.0061$$

$$\frac{\partial E}{\partial w_2^1} = -0.0122$$



Question 1

d) Calculate the output to the input values x_1 and x_2

1. Calculate $f(a^1)$

$$a^1 = 1 \times 1.0006 + 2 \times 0.1012 = 1.203$$

$$f(a^1) = 0.5 \times 1.203 = 0.6015$$

2. Calculate $f(a^2)$

$$a^2 = 0.6015 \times 0.1073 = 0.0645$$

$$f(a^2) = \frac{1}{1+e^{-0.0645}} = 0.5161$$

Param	Value
x_1	1
x_2	2
d	1
η (LR)	0.1
a^1	0.6
a^2	0.5150

Weight	Value
w_1^1	1.0006
w_2^1	0.1012
w^2	0.1242

$$a^1 = \phi_1(x) = 0.5x$$

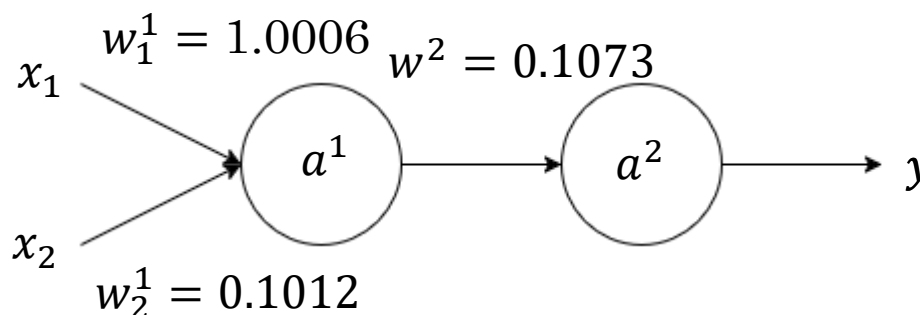
$$a^2 = \phi_2(x) = \frac{1}{1+e^{-x}}$$

$$y = a^2$$

$$d = 1$$

Q: What does this tell you?

A: Our previous prediction was 0.5150, which is less than our newly predicted value (0.5161). This means that after weight update, our model's error is less, therefore the prediction is more accurate.



Question 2

- Let's consider a 2-layer neural network
- 2 inputs: x_1 and x_2 with one output y .
 - $x_1 = 1$
 - $x_2 = 2$
- Activation functions (sigmoid and ReLU, respectively).
 - $\phi_1(x) = \frac{1}{1+e^{-x}}$
 - $\phi_2(x) = \begin{cases} x, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$
- This time, our hidden layer contains 2 neurons
 - Hidden layer activation function: $\phi_1(x)$
 - Output activation function: $\phi_2(x)$

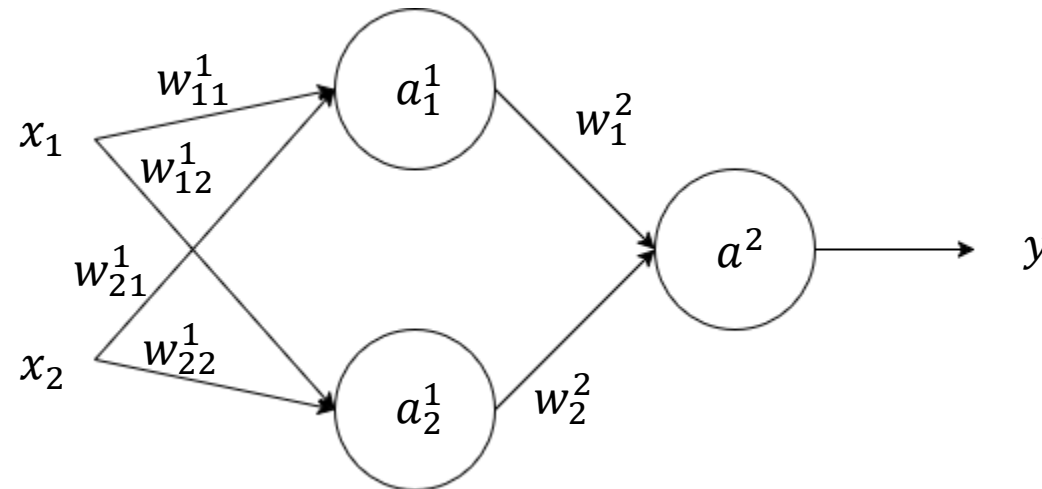
Question 2

- a) For each neuron, calculate its induced local field and output for the input values.
- b) Use backprop to compute the gradients of the error with respect to all weights.
- c) Calculate the updated weights for $\eta = 0.1$

Param	Value
x_1	1
x_2	2
d	0.5
η (LR)	0.1

Weight	Value
w_{11}^1	1
w_{12}^1	0.1
w_{21}^1	0.3
w_{22}^1	0.4
w_1^2	0.1
w_2^2	0.2

$$\phi_1(x) = \frac{1}{1+e^{-x}}$$
$$\phi_2(x) = \begin{cases} x, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$$



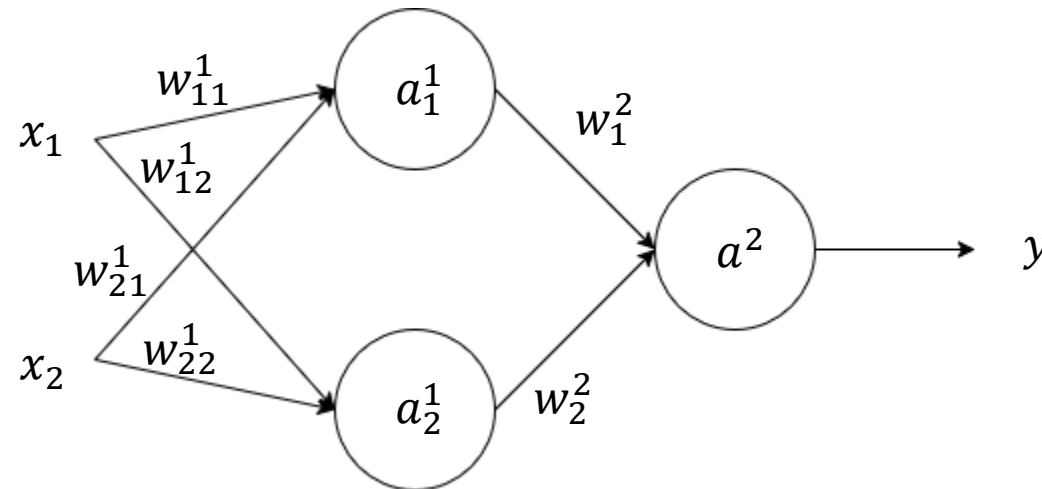
Question 2

a) For each neuron, calculate its induced local field and output for the input values.

1. $a_1^1 = 0.832$
2. $a_2^1 = 0.7109$
3. $a^2 = 0.2254$

Param	Value
x_1	1
x_2	2
d	0.5
η (LR)	0.1

Weight	Value
w_{11}^1	1
w_{12}^1	0.1
w_{21}^1	0.3
w_{22}^1	0.4
w_1^2	0.1
w_2^2	0.2



Question 2

b) Use backprop to compute the gradients of the error with respect to all weights.

1. Derivative of neuron a^2 :

$$e = 0.5 - 0.2254 = 0.2746$$

$$\phi'_2(a^2) = 1$$

$$\delta^2 = 0.2746 \times 1 = 0.2746$$

2. Hidden layer:

$$\delta_1^1 = 0.0038$$

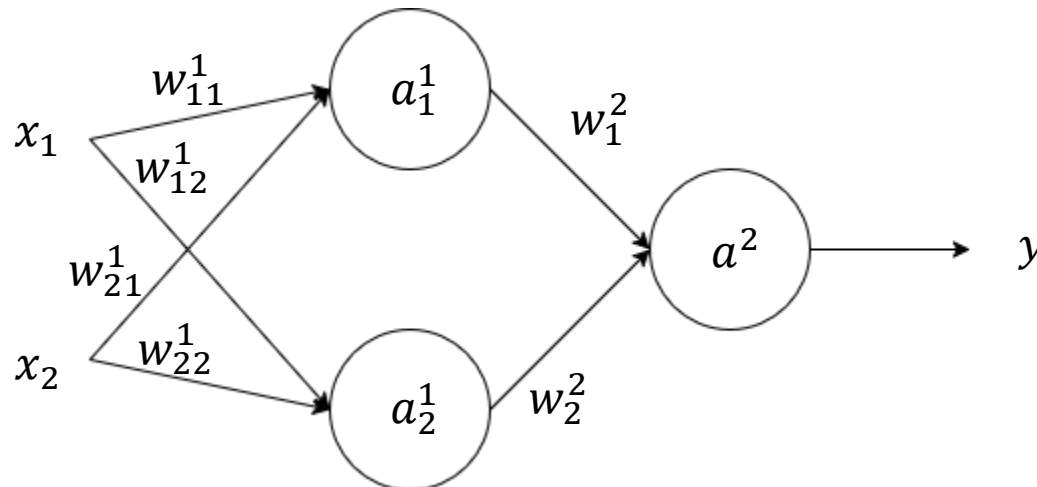
$$\delta_2^1 = 0.0113$$

3. Gradients:

$\frac{\partial E}{\partial w_1^2} = -0.2285$	$\frac{\partial E}{\partial w_{11}^1} = -0.0038$
$\frac{\partial E}{\partial w_2^2} = -0.1952$	$\frac{\partial E}{\partial w_{21}^1} = -0.0076$
	$\frac{\partial E}{\partial w_{12}^1} = -0.0113$
	$\frac{\partial E}{\partial w_{22}^1} = -0.0226$

Param	Value
x_1	1
x_2	2
d	0.5
η (LR)	0.1
a_1^1	0.832
a_2^1	0.7109
a^2	0.2254

Weight	Value
w_{11}^1	1
w_{12}^1	0.1
w_{21}^1	0.3
w_{22}^1	0.4
w_1^2	0.1
w_2^2	0.2



Question 2

c) Calculate the updated weights for $\eta = 0.1$

$\frac{\partial E}{\partial w_1^2} = -0.2285$	$\frac{\partial E}{\partial w_{11}^1} = -0.0038$
$\frac{\partial E}{\partial w_2^2} = -0.1952$	$\frac{\partial E}{\partial w_{21}^1} = -0.0076$
	$\frac{\partial E}{\partial w_{12}^1} = -0.0113$
	$\frac{\partial E}{\partial w_{22}^1} = -0.0226$

$w_1^{2,+} = 0.1229$	$w_{11}^{1,+} = 1.0004$
$w_2^{2,+} = 0.2195$	$w_{21}^{1,+} = 0.3008$
	$w_{12}^{1,+} = 0.1011$
	$w_{22}^{1,+} = 0.4023$

Param	Value
x_1	1
x_2	2
d	0.5
η (LR)	0.1
a_1^1	0.832
a_2^1	0.7109
a^2	0.2254

Weight	Value
w_{11}^1	1
w_{12}^1	0.1
w_{21}^1	0.3
w_{22}^1	0.4
w_1^2	0.1
w_2^2	0.2

